

Youtube

Functional requirements:

user should be able to open any videos based on his preference,
user should be able to download any videos offline based on his preference,
videos resolution should get adjusted based on network bandwidth

Non-functional requirements:

Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (eventual consistency is fine , but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:

user ,
video,
channel ,
feed

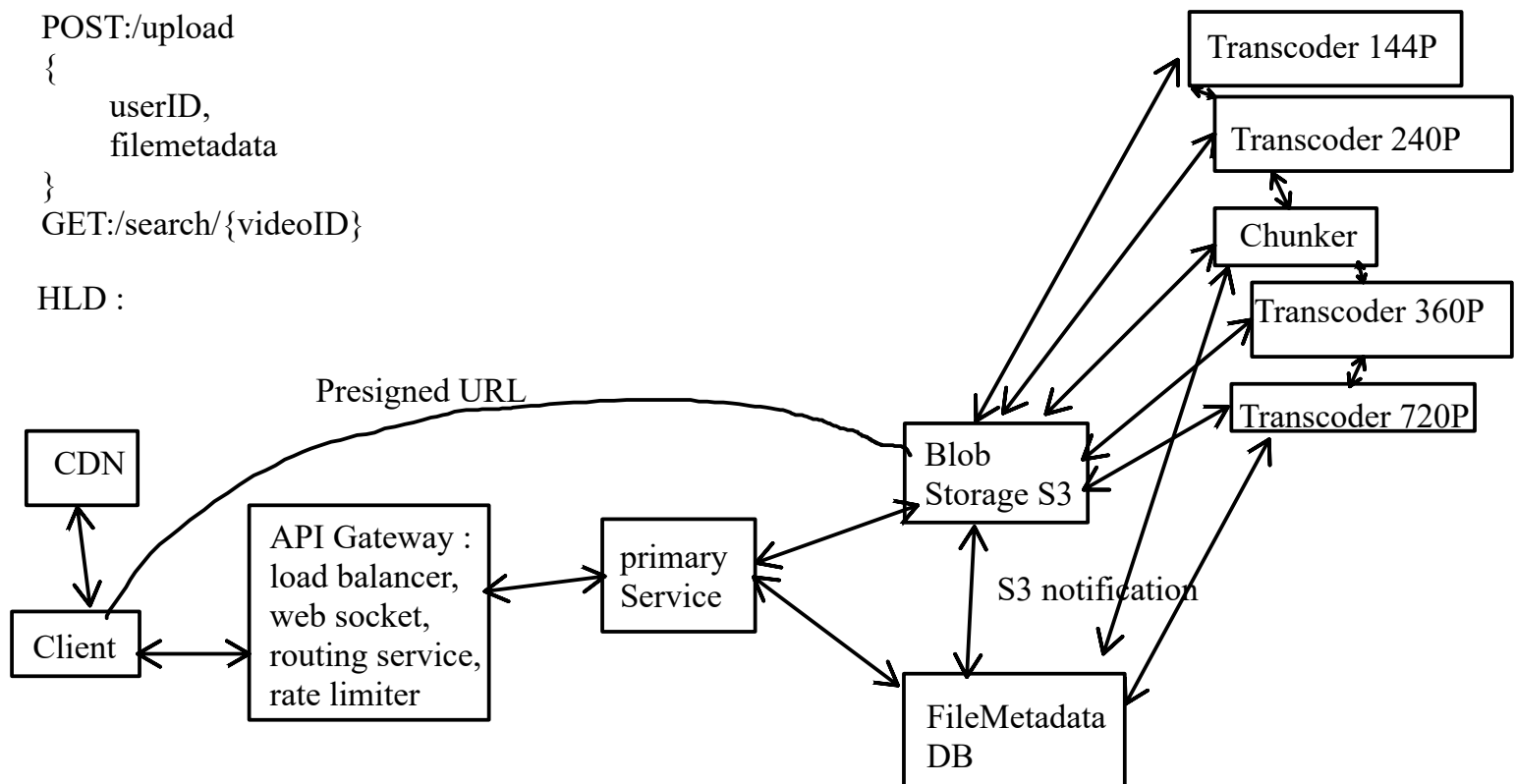
API :

POST:/upload

```
{
  userID,
  filemetadata
}
```

GET:/search/{videoID}

HLD :



presigned URL

1)